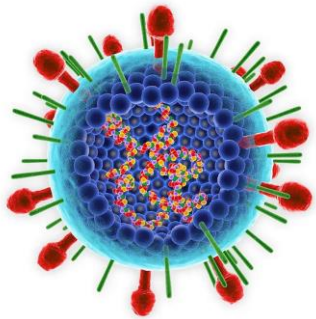


The Belgian National Survey on RSV pediatric burden of disease

Preliminary results

May 2025



Blumental S, De Mot L, Raes M
and the Belgian Pediatric RSV Study Group and the
BELSARI-net research group

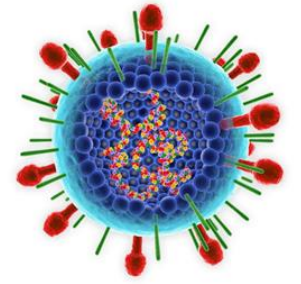


Setting



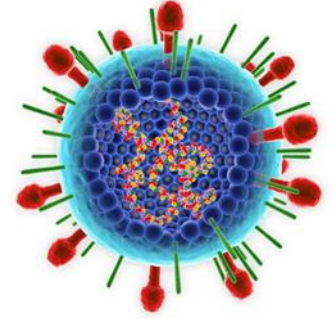
- ▶ **New preventive tools against RSV disease** in children available from 2024
Monoclonal antibodies (Beyfortus®)/Maternal vaccine (Abrysvo®)
 - ▶ **No specific surveillance for pediatric RSV burden until 2023-2024**
- Decision to build a **broad surveillance network** to better **assess the RSV pediatric burden of disease in Belgium (first in hospitals setting)** and to **monitor impact of preventive tools** once implemented
- ❖ **Adaptation of SARI (severe acute respiratory infection) hospital surveillance** to better catch pediatric RSV cases (since Nov 23)
 - ❖ **Creation of the RSVPED study**, run in parallel to SARI survey

Method



- ▶ New **common case definition** for pediatric severe acute respiratory infections
- ▶ **Questionnaire** adapted to pediatric features, **almost similar between both studies**
- ▶ **SARI surveillance (active since 2013)**
 - Inclusion of all cases hospitalized for severe acute respiratory infections matching the clinical case definition
 - Sample sent to the National Reference Center, tested for 18 respiratory viruses (PCR)
 - Weekly reporting (bulletin of acute respiratory infections)
- ▶ **RSVPED study**
 - Active recruitment of Belgian centers outside the SARI network
 - Inclusion of **all proven RSV** cases **0-4 years old** meeting the case definition and hospitalized during the enlarged epidemic period (October to March)
 - Retrospective **clinical data collection** by local team/data nurse
 - Preliminary **aggregated data** obtained, mostly from the **3 first months of the past season**

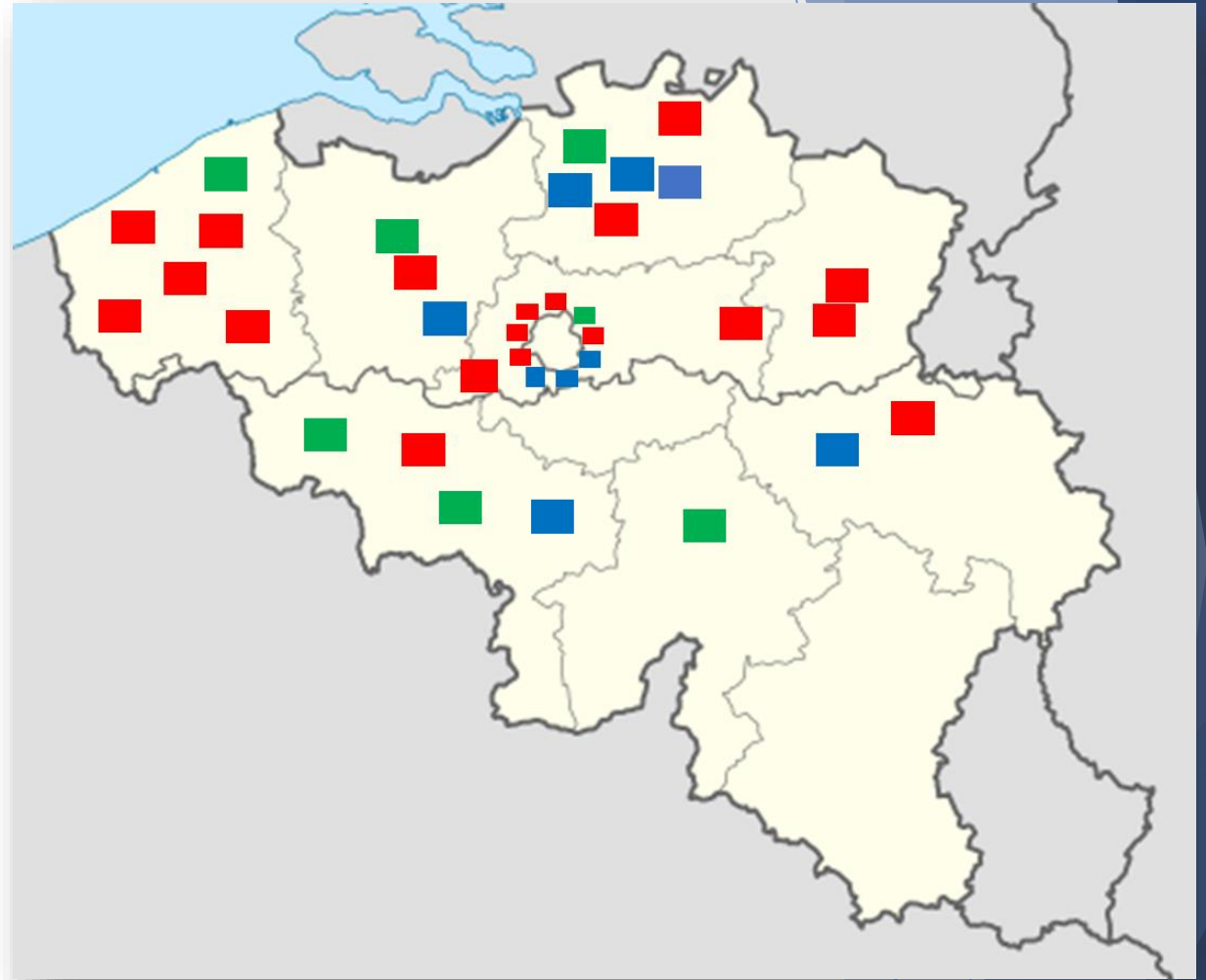
Timing



- ▶ **3 years of surveillance**
 - 2023-2024 (pre-prevention period);
 - 2024-2025 (first year of prevention implementation, one tool reimbursed),
 - Future 2025-2026 (second year of prevention, two tools reimbursed)
- ▶ **Merged of both datasets after each season and pooled analysis for common items**
- ▶ **Specific analyses leveraging main assets of each study**

34 participating sites all over the country

- ▶ **RSVPED : 27 centers**
 - 21 centers from which incidence data from both seasons are available (paired data)
 - 6 centers partial data collection
 - And more coming...
- ▶ **SARI : 7 additional centers**



UZ Brussels, UZ Gent, AZ Sint Jan Brugge, Centre Chwapi Tournai, CHU Namur-Godinnes, Grand Hôpital de Charleroi, ZAS ziekenhuizen, CHU St Pierre, CHU Tivoli, AZ Klina, AZ Turnhout, AZ Delta Roeselare, Jessa Ziekenhuis, Clinique St Jean, AZ St Maria Halle, AZ Jan Palfijn Gent, AZ Groeninge, Jan Yperman, Hôpital Delta Chirec, Hôpital St Anne-St Remy Chirec, Hôpital de Braine-l'alleul, HUDERF, OLV Wareghem, CHC Mont Leggia, AZ Diest, Imelda Ziekenhuis, Hôpital Iris Sud Ixelles, Cliniques Universitaires St Luc, Vitaz, Cliniques de L'Europe, Universiteit Ziekenhuis Antwerp, CHU/Citadelle Liège, AZ Blasius, CHU Charleroi

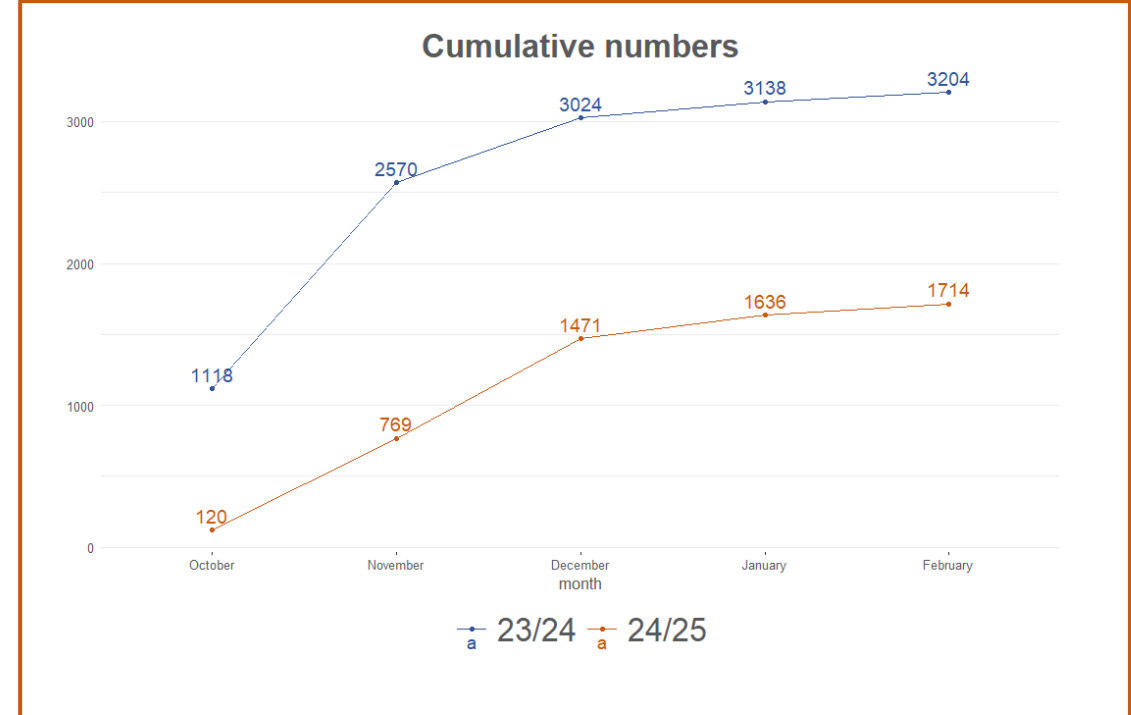
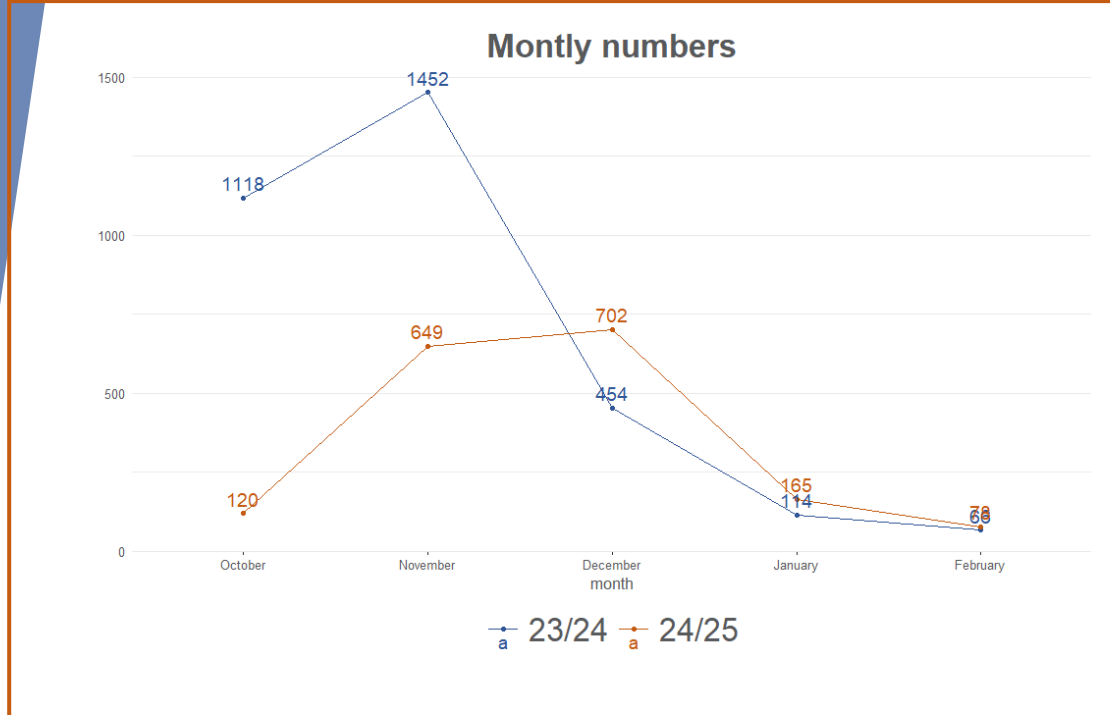
RSV: high pediatric burden of disease season 2023-2024

▶ RSVPED

- ▶ **>3000 cases** included from **23 centers** (even under estimated!)
 - ▶ Incidence: **16.3 hosp/1000 children** (0-4yo) between Oct and Feb
 - ▶ Estimated **~9500** pediatric hospitalizations between Oct and Feb
- ▶ 53,3% < 6 months old, 90% < 2 years old
- ▶ Median length of stay: 5 days
- ▶ Use of respiratory support in 61% of the cohort (67% in infants<6m)
- ▶ Antibiotics consumption in 34% of children (26% in infants <6months)
- ▶ 217 ICU admissions recorded from 5 PICU tertiary centers

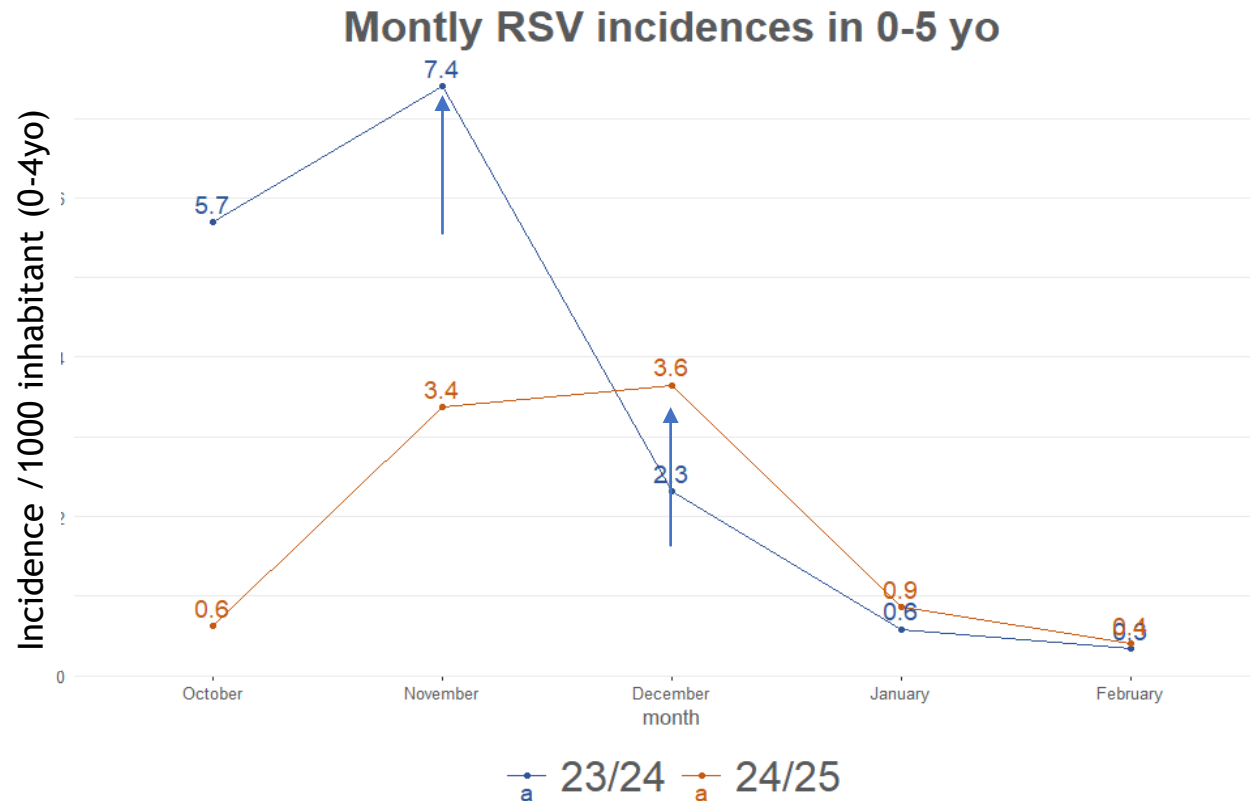


Number of children 0-4 y old hospitalized for RSV in 2024-2025 (first year prevention) compared to 2023-2024 (RSVPED)



- ❖ **Much less hospitalized cases** this season compared to the previous one
- ❖ **Shift in seasonality: last season started earlier** than in the pre-pandemic period (start in September in 2023, October in 2024)

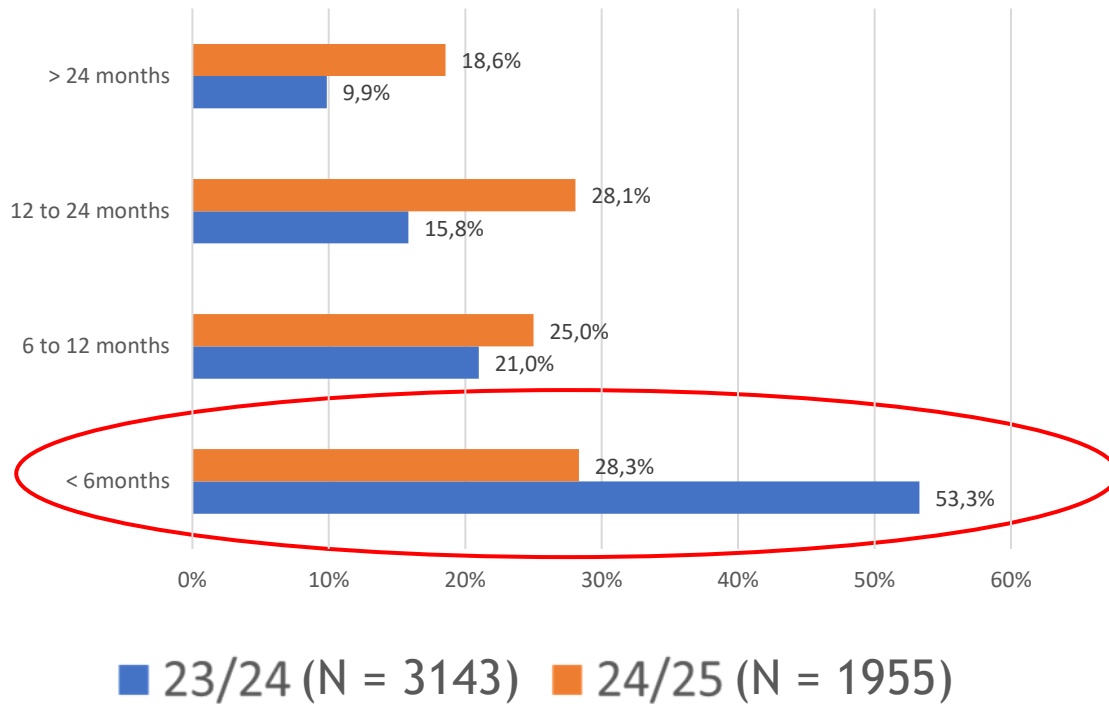
Incidence patterns of RSV hospitalizations in children 0-4 y old in 2024-2025 and 2023-2024 (RSVPED)



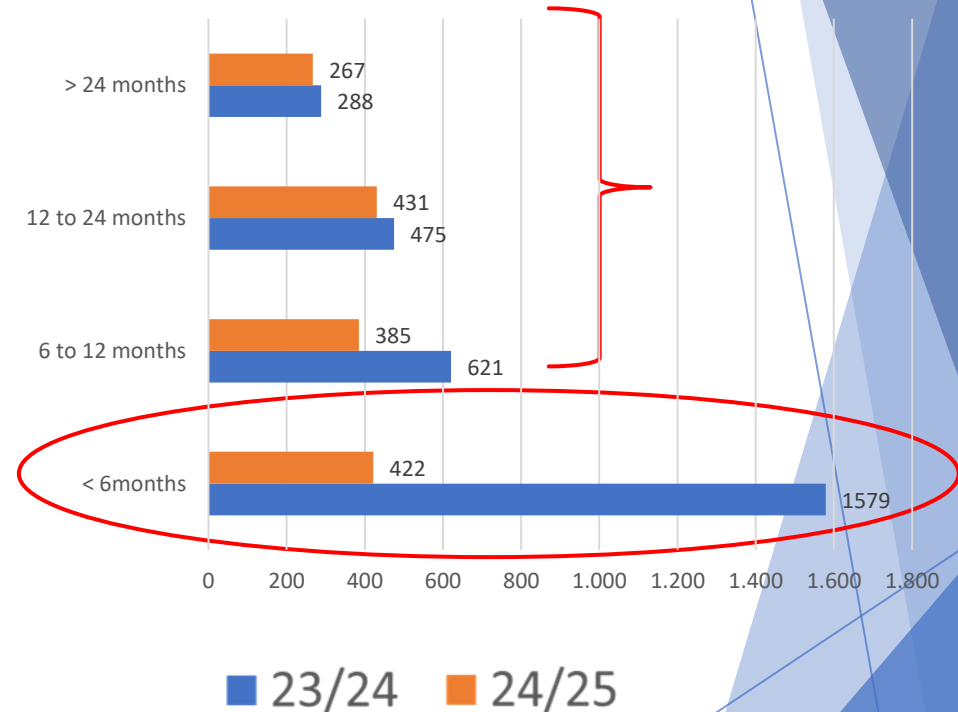
- ❖ **Estimation:** ~ **4300 fewer hospitalizations in Belgium** in 24-25 than in 23-24, between Oct and Feb (from which **> 2200 during month of the peak**)

Age distribution among RSV pediatric hospitalized cases during both seasons (RSVPED and SARI)

RSVPed + SARI

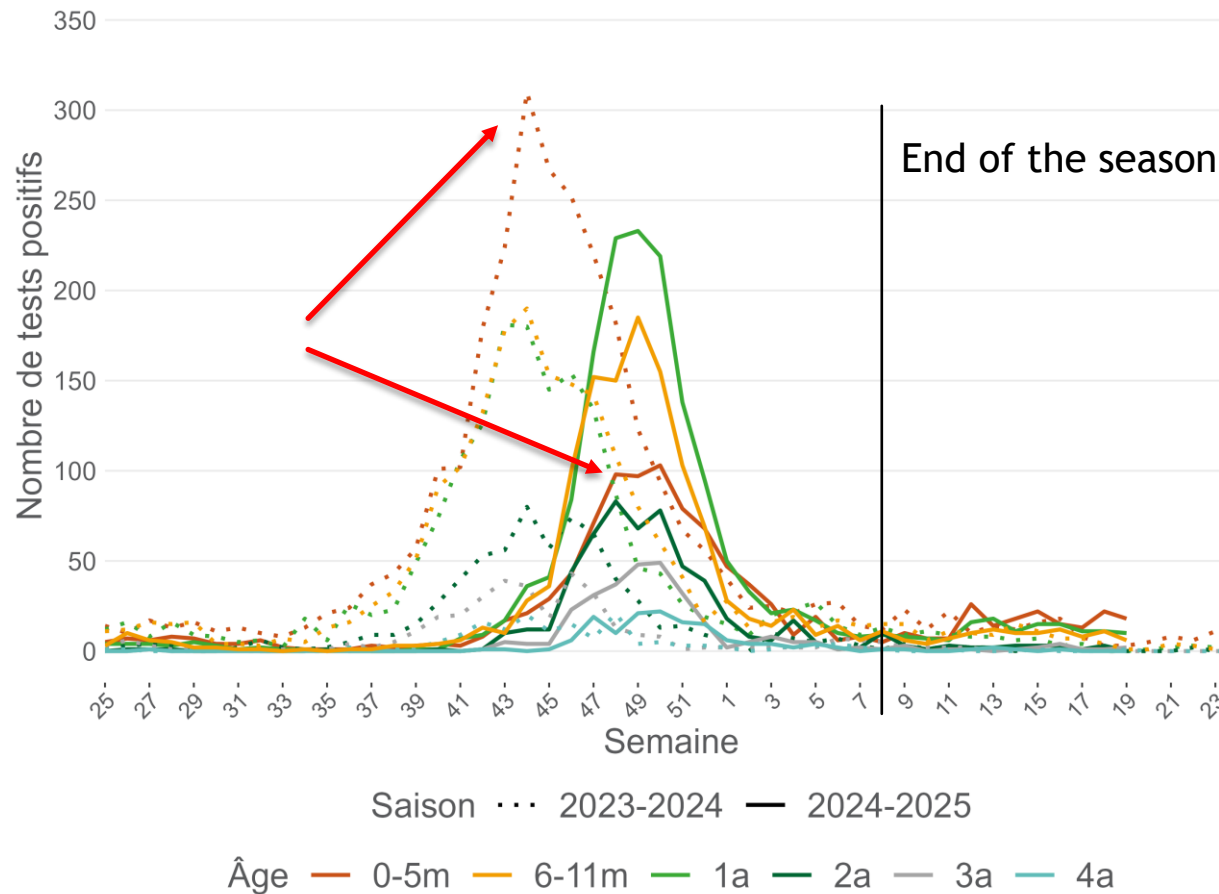


Absolute numbers (RSVPED)



Decrease of burden in <6 months in 2024-2025 (proportionally less represented)
Concordant results between both datasets
No changes in other age groups (no age shift)

Incidence pattern of ambulatory RSV cases in children 0-4y in 2023-2024 and 2024-2025 (sentinel labs, Sciensano)



Immunization with Nirsevimab among hospitalized infants eligible for prevention (2024-2025, N=270)

► 29/134 cases in SARI (**22%**)

(6 catch up program, 23 newborns)

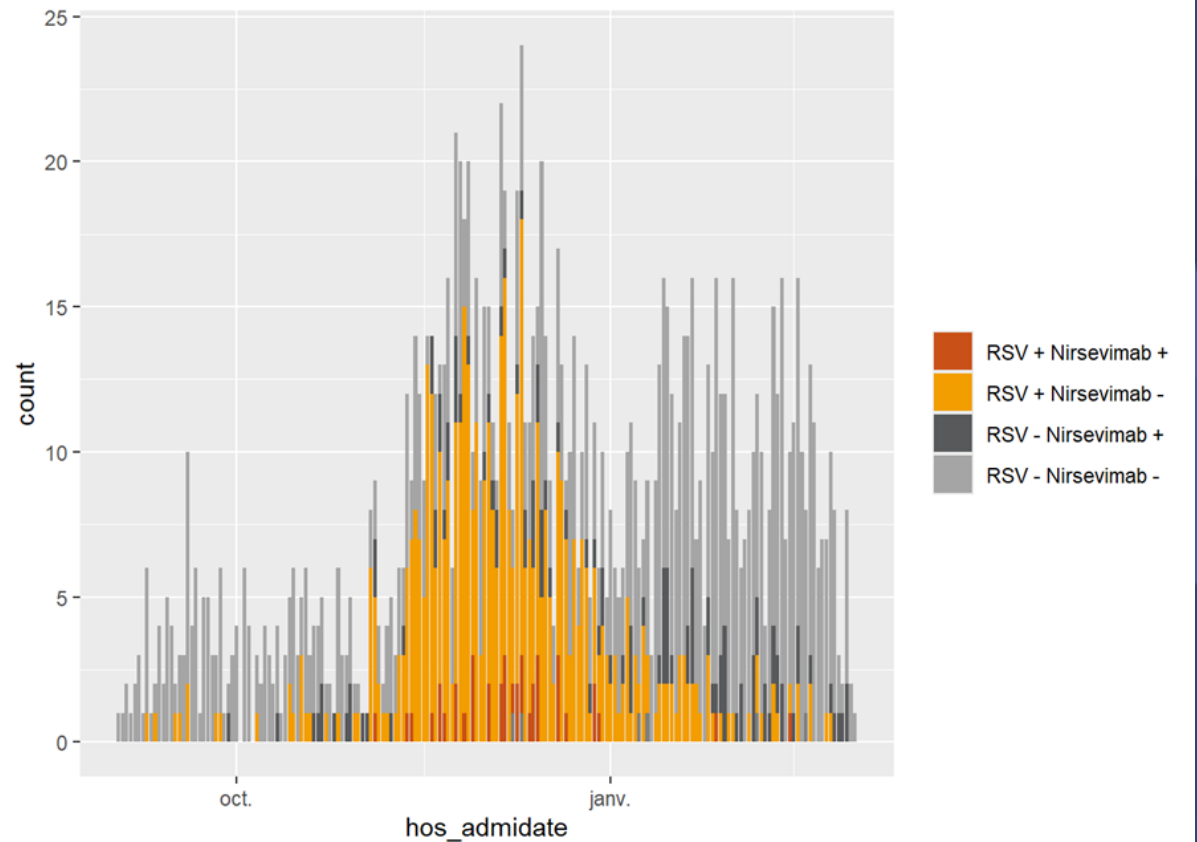
► 43/136 cases in RSVPed (**32%**)

(16 catch up program, 27 newborns)

(+ 3 with maternal vaccine)



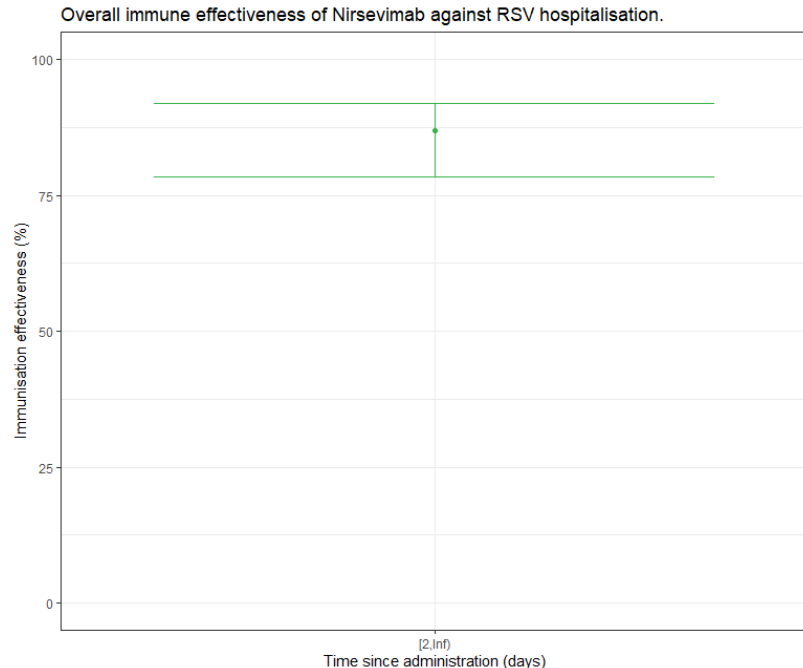
¾ of infants <6m hospitalized non immunized



Effectiveness of nirsevimab in preventing hospitalization (SARI, 2024-2025)

Test-negative design (common in vaccine studies):

- Compares nirsevimab-eligible SARI patients hospitalized because of RSV (**cases**) to those hospitalized for SARI but tested negative for RSV (**controls**).
- If nirsevimab is effective, fewer immunized infants should be found among RSV-positive cases than among RSV-negative controls

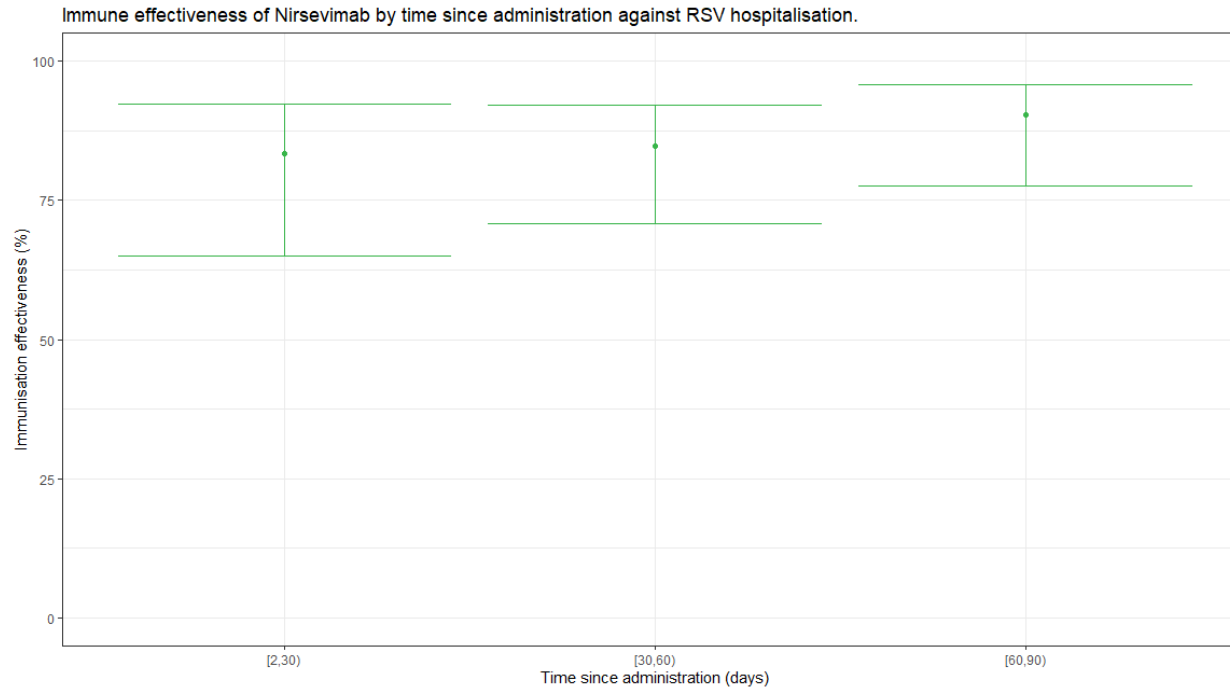


Immunization effectiveness = 87%

	RSV-negative SARI hosp	RSV-positive SARI hosp
Not immunized	121	140
Immunized	239	45

- Adjusting for confounders (through logistic regression): comorbidities, sex, sampling week, age in months
- Excluding nosocomial infections, children who received palivizumab or whose mother received Abrysvo

Effectiveness of nirsevimab in preventing hospitalization (SARI, 2024-2025)

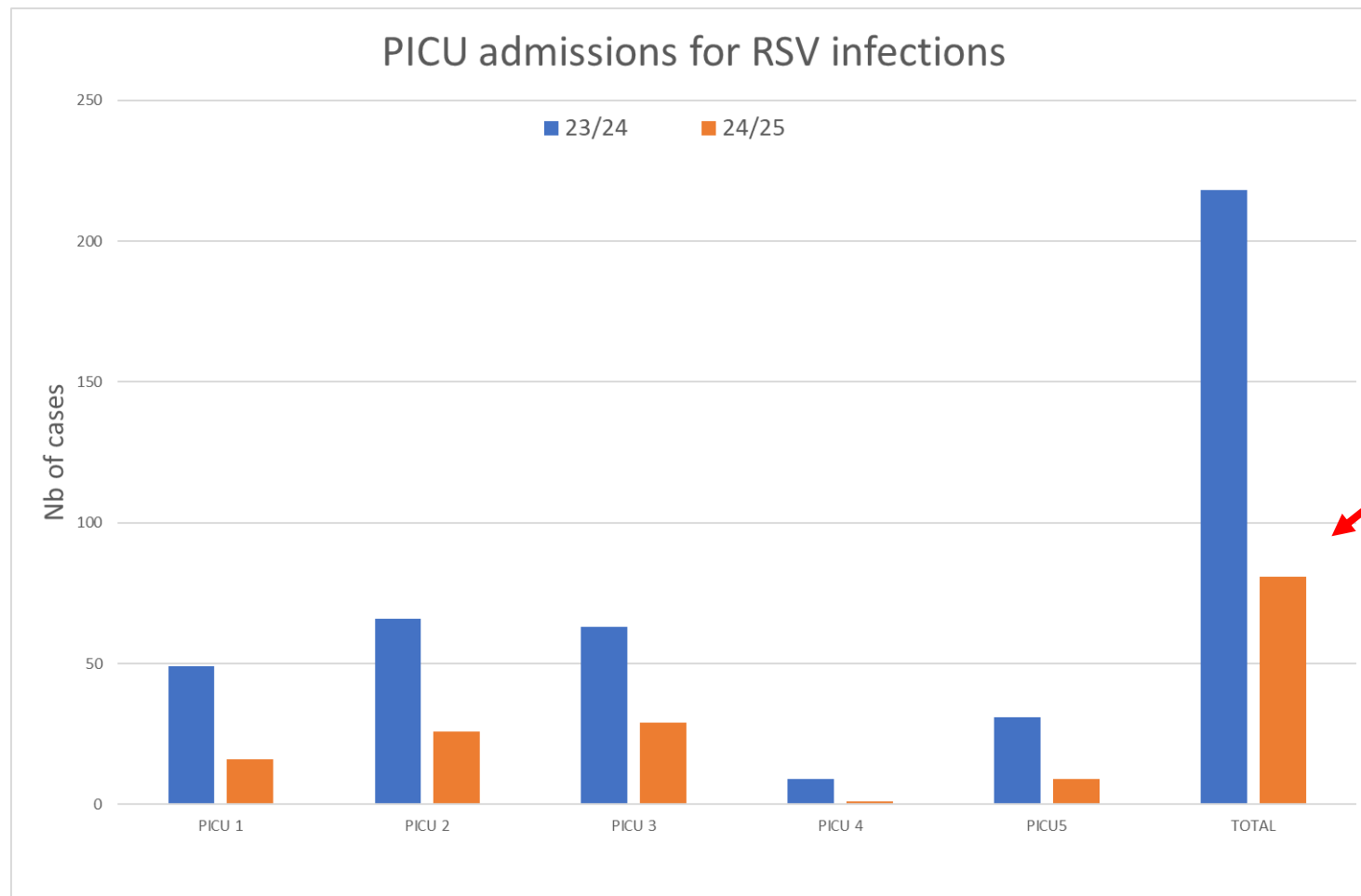


Treatment group	Number of SARI hospitalisations	Number of RSV positive hospitalisations	Immune effectiveness
Not immunised	261	140	/
Immunised 2 - 29 days ago	55	14	83.4% (95% CI: 65,0 - 92,2)
Immunised 30 - 59 days ago	90	20	84,7% (95% CI: 70,7 - 92,0)
Immunised 60 - 89 days ago	65	9	90,2% (95% CI: 77,6 - 95,7)

Effectiveness doesn't decrease until 90 days after administration

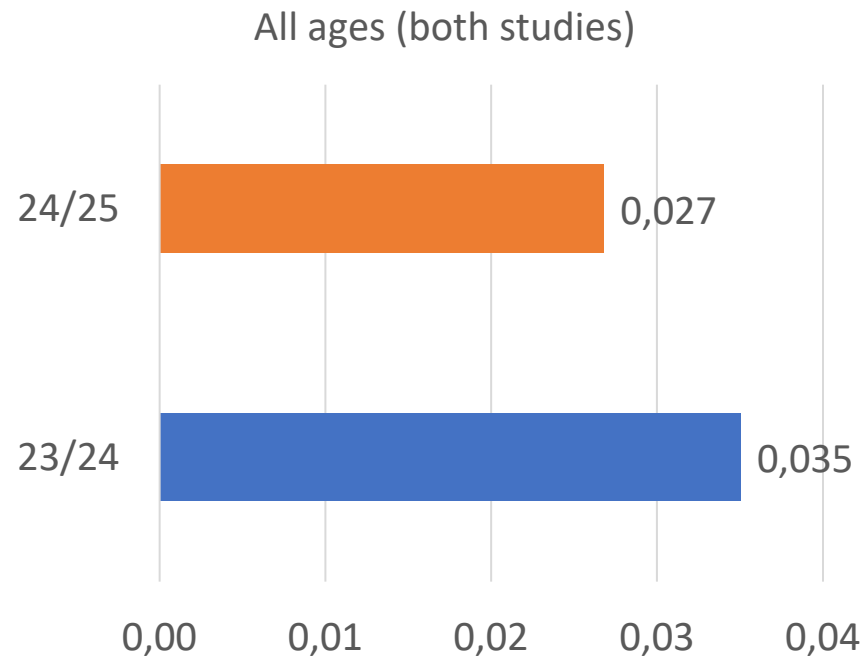
Severity of infections/burden of disease ICU admissions (both studies)

- Decrease of total ICU admissions (5 main PICU centers, 3 regions)



Severity of infections/burden of disease ICU admissions (both studies)

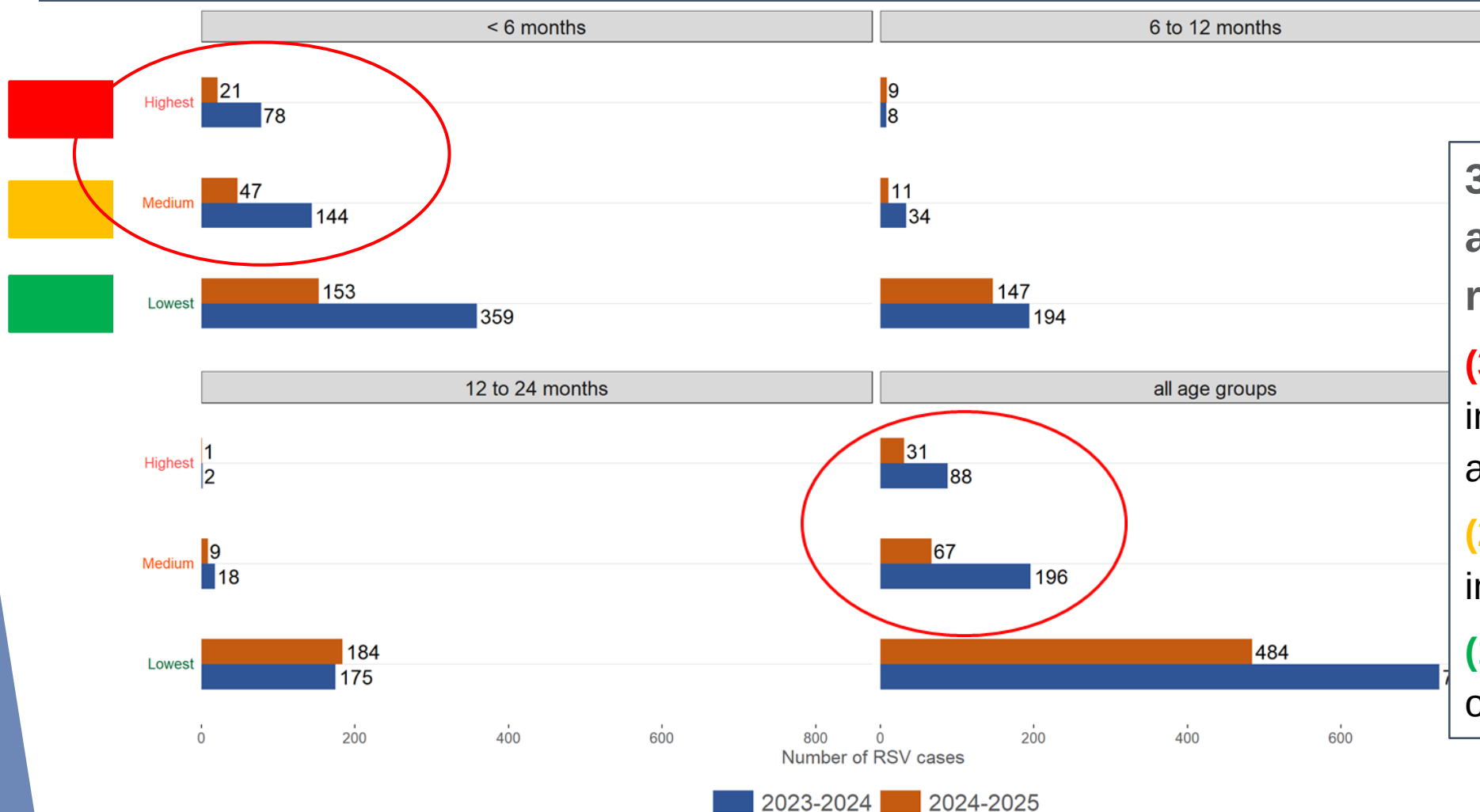
- Percentage of children transferred to PICU in tertiary hospitals amongst all admitted in primary hospitals



Fatality rate: 0% in both seasons

Burden of infection according to RSV seasons

Pediatric wards (both datasets, N=1594)



3 grades of severity according to use of respiratory support

(3) = highest severity: invasive ventilation or ICU admission

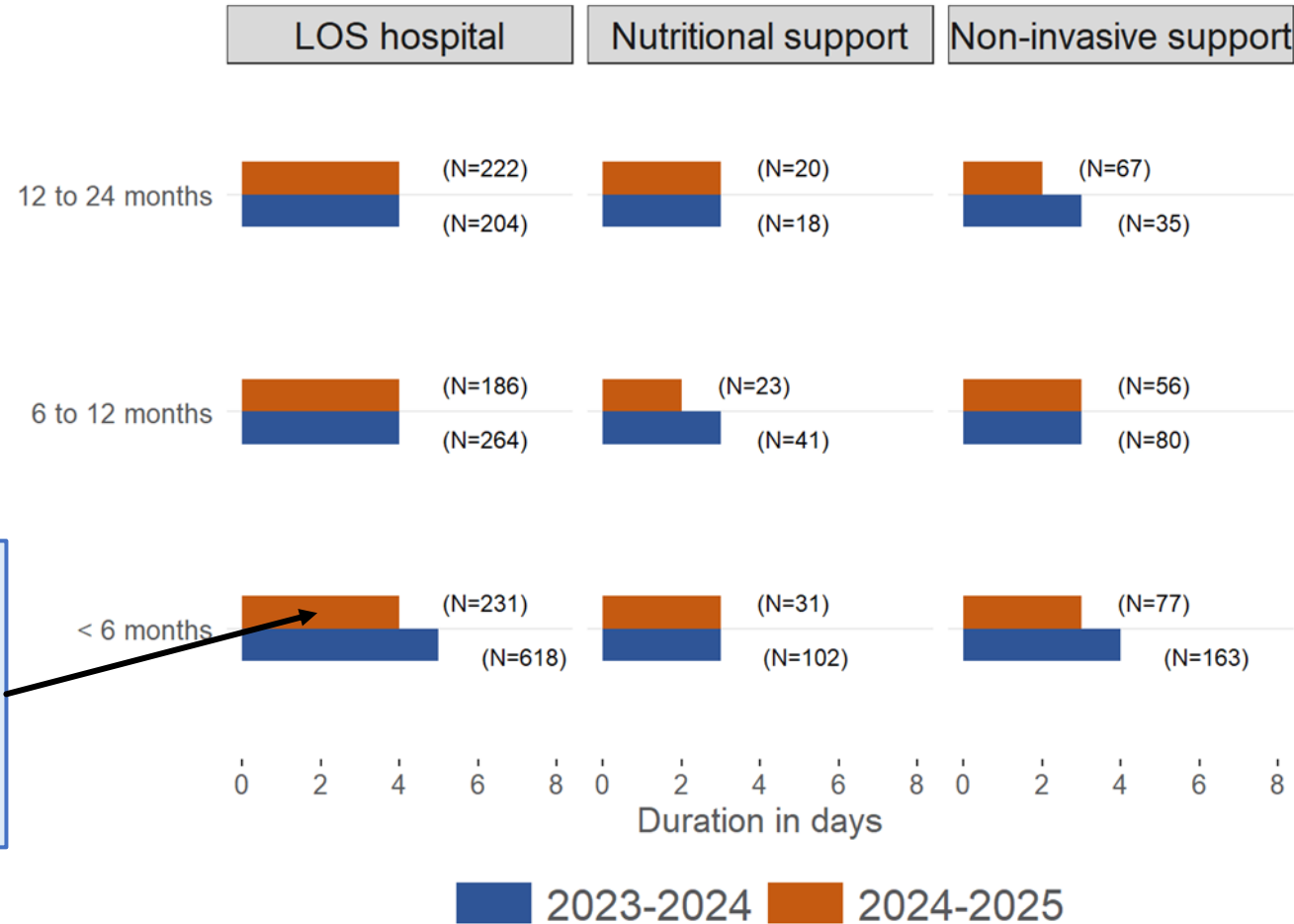
(2) = mid-severity: non-invasive ventilation

(1) = low-severity: none or oxygen goggles

Important reduction of infants requiring respiratory support !! (middle and high severity)
No increase of severity/burden in other age groups

Burden of infection according to RSV seasons

Pediatric wards (both datasets)



Median length of stay
in 2024-2025
(<6months old):
4 days vs 5 days in
2023-2024

Immunization with Nirsevimab among hospitalized infants eligible for prevention (2024-2025, N=270)

► 29/134 cases in SARI (**22%**)

(6 catch up program, 23 newborns)

► 43/136 cases in RSVPed (**32%**)

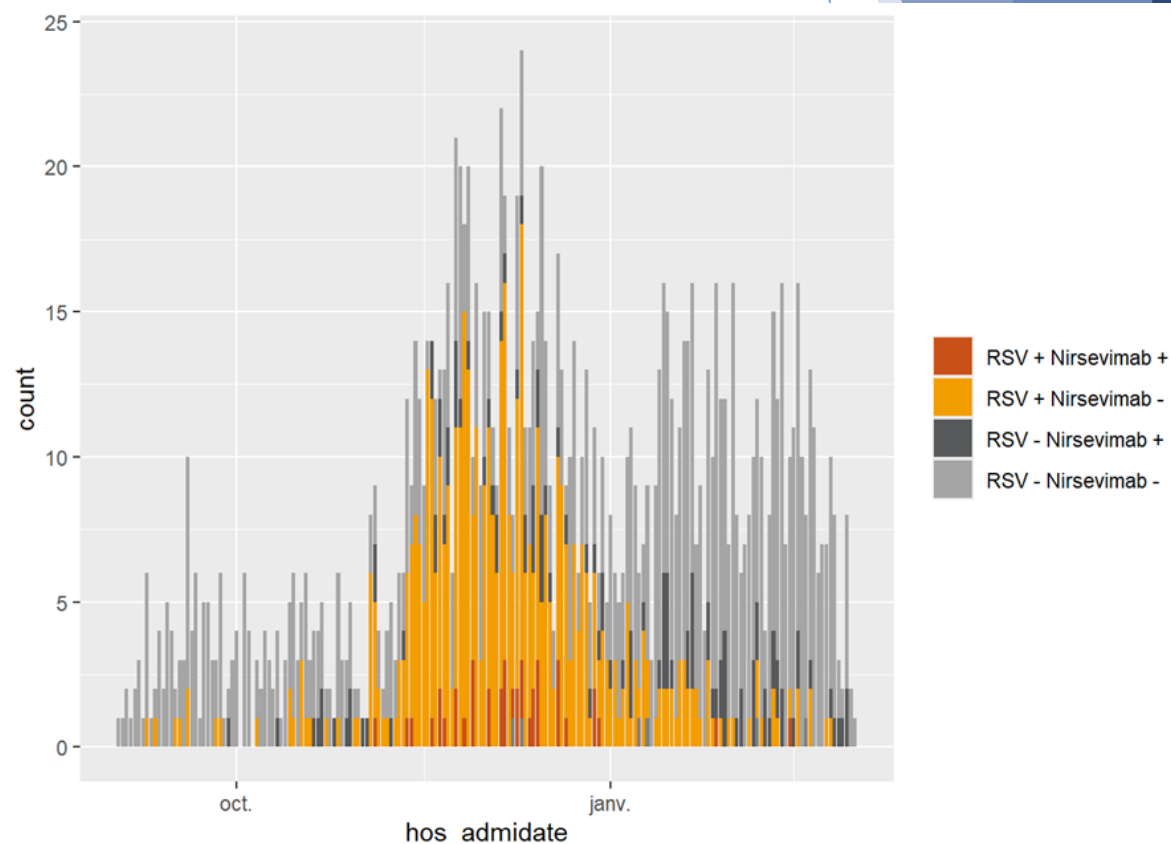
(16 catch up program, 27 newborns)

(+ 3 with maternal vaccine)



► Infants hospitalized and immunized with nirsevimab :

- represent **7,5 %** of the whole cohort of RSV hospitalized children 0-4y
- have a few **more risk factors** (**12,5% vs 6,1%** in non-immunized) than other in the same age group
- have a few **more prematurity** (**18,2% vs 6,1%** in non-immunized) than other in the same age group



What was the immunization uptake
for this first year of prevention?

Nirsevimab implementation in Belgium

30 hospitals - Pilot Study (Pr Raes)

- **Maternity** (25 hospitals from Flanders, Oct-Dec, same trend in Jan/Feb)

	Octobre	Novembre	Decembre
Naissances	3.546	2.755	2.714
Couverture	(62, 12%) – 86% – (96.82%)	(71.52%) – 81,75% – (100%)	(82.29%) – 84,25% – (100%)

- **Catch-up** (Octobre): 27 hospitals from Flanders/Brussels/Wallonia

9.580 inj / 27 hospitals → 35.500/100 hôpitaux/60.000 enfants/ 6 mois = **59%**

Injections/ invitations envoyées: 9.539/17.762 (**53,70%**)

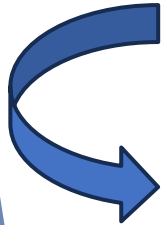
+ More data expected about ambulatory delivery (intego, farmanet)

If all eligible patients would have received Nirsevimab (100% uptake) this season (both studies)



Expectable proportion of hospitalizations that could have been avoided based on:

- The **proportion of admitted children eligible for immunization**: 39% (maternity 18%; catch up 21%)
 - The **percentage of non-immunized amongst eligible (73%): 61%** maternity, 85% catch up
 - The **average efficacy of Nirsevimab** to prevent hospitalization (literature data) : **88%**
- In 2024-2025



An extra **reduction of ~25%** of the total of RSV-related hospitalizations in children 0-4y could be expected if full uptake → at least reduction **~1000-1300 pediatric hospitalizations/year**

Why was the catch-up uptake lower than expected??

Acceptability of RSV prevention to Parents and Health Care Workers

Blumental S and Grossman D, Belgian Journal of Pediatrics Decembre 2024

- ▶ Multicenter opinion survey among young parents and antenatal caregivers in Brussels

→ **New acceptability study in preparation**

- ▶ Broader recruitment of parents:
Chirec Network, UZ Ghent, Jessa ziekenhuis, Chwapi Tournai + ONE, Kaleido, (k&G?)
- ▶ + Assessment of opinion of pediatricians, gynecologists and midwives all over the country
- ▶ Goals: identify expectancies toward prevention
+ strengths and caveats of the current campaign
- ▶ The COVID-19 pandemic reduced the confidence of 2.1% of parents in vaccination

Take home messages

Promising results but we can do better!



- ▶ From Oct to end February: **reduction of RSV pediatric hospitalizations** in 2024-2025 (first season with RSV prevention) compared to 2023-2024
- ▶ **Highest impact** seen among the **< 6months old** (significant decrease of number of cases and complications)
- ▶ **Reduction of RSV PICU burden**
- ▶ **Sub-optimal uptake** of the preventive tool in the catch-up cohort (better in maternity)

Efforts should be done to:

- **Identify expectancies** and feelings of **parents and health care workers** about RSV prevention
- **Improve communication** about the campaign through various channels
- **Alleviate administrative workload** linked to the preventive campaign

Perspectives



- ▶ **In-depth analysis and comparison between both seasons** of pattern of RSV infection, antibiotics consumption, coinfections rate and complications.
- ▶ **Further comparison between** pattern of infection of **immunized and non-immunized** infants
- ▶ **Third year of study: some changes for the new upcoming season:**
 - Two tools reimbursed: nirsevimab and maternal vaccine
Uptake and impact of both tools should be assessed together and separately
 - Expected better uptake
Better communication, changes in reimbursement criteria, better awareness of parents and HCWs
 - Request submitted for reimbursement of nirsevimab for children with comorbidities
(for their second RSV season, decision pending)
- ▶ **Cost effectiveness analysis from KCE ongoing**
- ▶ **Genomic characterization of RSV in breakthrough infections and monitoring over time (NRC respiratory pathogens)**
- ▶ **Future new tools coming soon**
(new monoclonals : clesrovimab, vaccines for older children)

Acknowledgements

- ▶ The team surveillance of respiratory infections (EID), Sciensano

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 - ▶ Nathalie Bossuyt (Sciensano)
-
- ▶ The Care Fundation Chirec



